

AI Beauty Contest

LLM-Driven Mission Control for ROS 2 Robotic Inspection

Simulation-First Inspection, Patrolling and Monitoring

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Abstract

This report presents a simulation-first mission-control and evidence platform for Unitree Go2 quadruped robots. The system enables an operator to express an inspection, patrolling or monitoring objective in natural language. A repository-local large language model decomposes the objective into a constrained set of robotic tasks, while deterministic planning and safety components validate and execute those tasks through ROS 2.

The prototype integrates natural-language mission planning, guarded navigation, single-robot and four-robot coordination, camera and LiDAR visualization, evidence collection, report generation and deployment history within a unified browser interface. The report describes the initiative and its value, the solution architecture, the implemented robotic skill, the measured key performance indicators and the associated business case. Results are presented as simulation evidence and are not interpreted as proof of physical-robot safety or field performance.

Contents

Abstract	i
1 Initiative Description and Value	1
1.1 Problem and Selected Use Case	1
1.2 Target Customer and Operational Need	1
1.3 Proposed Initiative	2
1.4 Customer Value and Differentiation	3
1.4.1 Value Proposition	3
1.4.2 Differentiation	3
1.5 Value Measurement	3
1.6 Scalability and Reusability	5
1.7 Current Maturity and Evidence Boundary	5
1.8 Success Definition and Next Validation Step	6
2 Solution Architecture	7
2.1 Architectural Principle	7
2.2 System Overview	7
2.3 Component Responsibilities and Traceability	7
2.4 Mission Processing and Authority	7
2.5 Principal ROS 2 Interfaces	10
2.6 Safety and Failure Handling	10
2.7 Design Rationale	11
2.8 Architectural Boundaries	11
3 Documented and Working Robotic Skill	12
3.1 Skill Definition and Purpose	12
3.2 Skill Contract	12
3.3 Preconditions	13
3.4 Execution Procedure	13
3.5 Skill State Model	14
3.6 Postconditions	14
3.7 Safety Constraints	14
3.8 Implementation and Verification	14
3.9 Representative Execution	15
4 End-to-End Demonstration	17
4.1 Choose an Operation	17
4.2 Platform Workspace	17
4.3 Create a Mission in Natural Language	19
4.4 Supervise Navigation and Inspection	19
4.5 Review Evidence, Not Only a Chat Answer	21
4.6 Operate a Four-Robot Fleet	21

4.7	Select and Reuse Configured Deployments	24
4.8	End-to-End User Flow	25
5	KPIs and Results	27
5.1	Evaluation Scope and Evidence Source	27
5.2	KPI Definitions and Denominator Rules	27
5.3	Planned Nominal Mission Matrix	28
5.4	Recorded Mission Results	28
5.5	Human Intervention	29
5.6	Perception False Positives and False Negatives	29
5.7	Secondary Observations	30
5.8	Result Status and Next Measurement Gate	30
6	Business Case	32
6.1	Scope and Commercial Recommendation	32
6.2	Initial Customer and Operational Problem	32
6.3	Proposed Commercial Offer	33
6.4	Cost and Pricing Assumptions	33
6.5	Expected Revenue Scenarios	34
6.6	Illustrative Customer Value	34
6.7	Scalability	35
6.8	Risks and Decision Gates	35
6.9	Business-Case Conclusion	35

Initiative Description and Value

1.1 Problem and Selected Use Case

Warehouse and logistics facilities require recurring inspection, patrolling and monitoring of receiving docks, storage zones, loading areas and operational equipment. These activities are necessary for detecting misplaced assets, obstructions, damaged equipment and other visible anomalies, but they can be repetitive, time-consuming and difficult to document consistently. Some areas may also expose personnel to moving equipment, poor visibility or other operational hazards.

Mobile robots can perform part of this work, but requesting a new mission typically requires knowledge of ROS 2, navigation interfaces, semantic maps, sensor configuration and robot-specific commands. An operational objective such as

“Inspect the receiving dock for barrels, save the evidence and produce a findings report.”

may therefore need to be translated manually into routes, observation points, camera actions and reporting steps by a robotics engineer. This creates a gap between the person who understands the operational need and the technical system capable of executing it.

The selected use case is **supervised warehouse inspection**. It is a bounded instance of the inspection, patrolling and monitoring applications defined by the challenge. The objective is not to give a language model direct control of a robot. The objective is to allow an operator to express the desired outcome in natural language and have the system convert it into validated, observable and interruptible ROS 2 operations.

The initial customer hypothesis is a warehouse or logistics operator that performs recurring visual inspections and needs a consistent record of the result. The software is intended to complement, rather than replace, existing robotics, safety and operational procedures.

1.2 Target Customer and Operational Need

The initiative distinguishes the user of the application from the people who purchase, approve and integrate it. Their principal needs are summarized in [Table 1.1](#).

The primary operational need can therefore be expressed as follows:

Enable a non-robotics operator to request, supervise and document a bounded warehouse inspection while deterministic software retains authority over physical execution and safety-related decisions.

This need is commercially relevant only if the resulting workflow proves more usable, repeatable or economical than the customer’s existing process. The current prototype demonstrates the technical workflow in simulation; customer baseline measurements and physical validation remain part of the proposed pilot stage.

Table 1.1: Target stakeholders and the needs addressed by the initiative.

Stakeholder	Operational need	Application response
Operations, maintenance or safety operator	Request and supervise an inspection without constructing ROS 2 commands or robot-specific routes.	Natural-language mission requests, visible task progress, camera and LiDAR views, evidence review and independent human controls.
Warehouse or facility manager	Obtain repeatable inspection coverage and evidence associated with a known mission, location and time.	Validated semantic tasks, configured viewpoints, mission history and evidence-based findings reports.
Automation or robotics lead	Introduce new missions without allowing generative output to bypass navigation and safety constraints.	Strict task schemas, deterministic route planning, bounded execution and explicit readiness checks.
Robotics systems integrator	Configure the solution for another site, robot profile or inspection workflow without rewriting the complete application.	Configuration-driven maps, rooms, inspection viewpoints, recognition targets, robot profiles and launch parameters.

1.3 Proposed Initiative

The initiative is an LLM-driven mission-control and evidence platform for Unitree Go2 quadruped robots, implemented as two ROS 2 packages and validated first in MuJoCo simulation. It combines the following capabilities:

- natural-language interpretation using a repository-local Qwen3-4B model executed through llama.cpp;
- deterministic fallback handling for supported instructions;
- a strict mission schema containing predefined navigation, patrol, capture, inspection, return and reporting tasks;
- semantic resolution of named rooms and operational areas;
- deterministic route planning and guarded goal execution;
- independent Stop, cancellation, manual takeover and recovery paths;
- camera and LiDAR visualization through a browser interface;
- evidence collection with mission, robot, task, location, viewpoint, timestamp and pose identity;
- findings-report generation when explicitly requested;
- single-robot operation and scenario-bounded coordination of four simulated robots.

The language model interprets the operator’s objective and proposes semantic work. It cannot publish actuator commands, invent physical coordinates, generate executable code, choose arbitrary filesystem paths or bypass configured mission constraints. Deterministic components validate the proposal, resolve the physical details and decide whether execution is permitted.

During execution, the operator supervises the mission through a unified browser workspace containing the semantic map, three-dimensional scene, live camera feeds, LiDAR visualization, task state, mission progress and collected evidence. The application therefore covers the complete simulated workflow from operational request to documented result.

The principal contribution is therefore not the language model in isolation. It is the controlled integration of natural-language interpretation, deterministic mission planning, guarded ROS 2 execution, human authority and evidence management within one observable workflow.

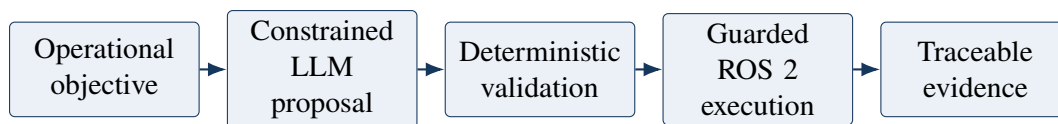


Figure 1.1: Transformation of an operational objective into a bounded, executable and traceable robotic mission.

1.4 Customer Value and Differentiation

1.4.1 Value Proposition

The proposed value is a reduction in the technical effort required to request, supervise and document a robotic inspection. Operators communicate at the level of operational objectives, rooms and assets, while the platform converts those objectives into validated robot tasks.

The resulting value proposition is:

Turn a warehouse inspection objective expressed in natural language into a guarded and traceable ROS 2 mission, without requiring the operator to write robot commands and without granting the language model physical control authority.

The initiative is expected to create value in four areas:

- **Accessibility:** operational personnel can request supported missions without constructing ROS 2 command sequences;
- **Repeatability:** inspections use configured routes, viewpoints and task contracts rather than an informal sequence of actions;
- **Human authority:** Stop, cancellation, manual takeover and recovery remain independent of the language model;
- **Traceability:** evidence is associated with the mission, robot, task, location, viewpoint, timestamp and measured pose that produced it.

1.4.2 Differentiation

Table 1.2 distinguishes the proposed initiative from representative alternatives.

Repository-local language and vision models also reduce dependence on public inference services. This supports future on-premises evaluation where facility imagery, maps and operational instructions should remain within the deployment environment. This is a deployment characteristic of the prototype, not yet a validated customer purchasing requirement.

1.5 Value Measurement

The customer value has not yet been measured at a physical site. To avoid presenting hypotheses as established outcomes, the initiative defines how each proposed benefit will be tested.

The challenge results for mission completion, human intervention and visual classification are reported in Chapter 5. Customer labour savings, return on investment and physical-site performance remain future pilot measurements.

Table 1.2: Differentiation from representative inspection approaches.

Approach	Typical limitation	Proposed differentiation
Manual inspection	Requires a person to perform and document every patrol directly.	Supports repeatable simulated robot missions with automatically associated evidence while retaining human supervision.
Fixed scripted robot mission	A changed objective may require an engineer to edit or trigger a different predefined sequence.	Natural-language interpretation can compose supported semantic tasks without changing the physical execution interfaces.
Engineering-led ROS 2 control	Exposes technically powerful interfaces but requires robotics knowledge from the mission author.	Presents rooms, assets and operational objectives through an operator-facing mission workspace.
Unrestricted LLM-to-robot control	Generative output may influence physical execution without a sufficiently narrow authority boundary.	Restricts the model to semantic proposals and keeps coordinates, routing, safety checks and execution deterministic.
Disconnected evidence workflow	Images and findings may require manual association with the mission that produced them.	Preserves evidence provenance across runtime, location, mission, robot, task, viewpoint, timestamp and pose.

Table 1.3: Value hypotheses and their corresponding evaluation measures.

Value hypothesis	Evaluation measure	Required comparison
Easier mission creation	Time and operator actions required to request an accepted mission.	Natural-language workflow compared with the existing engineering-led procedure.
Reliable mission execution	Mission completion rate and bounded failure reasons.	Repeated execution of a predefined mission suite under documented conditions.
Reduced supervision burden	Human intervention rate and intervention type.	Number of interventions divided by attempted missions, with manual test interventions reported separately.
Reliable visible findings	False-positive and false-negative rates for the selected targets.	Qualified perception output compared with labelled ground truth.
Complete inspection evidence	Proportion of required viewpoints with correctly attributed accepted evidence.	Expected viewpoint set compared with stored evidence records.
Faster documentation	Time from mission completion to an available findings report.	Proposed evidence workflow compared with the current reporting procedure.

1.6 Scalability and Reusability

The platform is designed around a reusable mission-control core and configuration-driven deployment data. The reusable core contains the LLM interface, task schema, mission state model, task compiler, routing and safety logic, ROS 2 integration, operator interface, perception workflow and evidence pipeline.

Site-specific configuration defines:

- semantic maps and traversable regions;
- named rooms, zones and aliases;
- patrol seeds and inspection viewpoints;
- recognition targets and mission presets;
- scene assets and terrain information;
- robot, sensor and locomotion parameters.

This separation is important because a scalable solution cannot require the entire mission-control application to be rewritten for every site. New environments still require mapping, inspection design and safety validation, but they can reuse the same mission vocabulary and execution pipeline.

Table 1.4: Implemented scalability mechanisms and their current evidence.

Dimension	Implemented mechanism	Current evidence and boundary
Mission reuse	Shared semantic task schema and composite inspection workflow.	The same task concepts support navigation, patrol, capture, inspection, return and reporting requests.
Environment reuse	Configuration-driven maps, rooms, routes, viewpoints and targets.	Demonstrated across warehouse, agricultural and scanned real-location scenarios in the repository.
Robot configuration	Robot and locomotion profiles are resolved independently from the natural-language task model.	Multiple repository-bundled locomotion profiles are selectable, but physical robot portability is not yet demonstrated.
Fleet extension	Robot-specific namespaces, task allocation, room ownership and route coordination.	Demonstrated with four robots in one bounded MuJoCo warehouse scenario; production fleet scale is not established.
Model deployment	Repository-local, hash-verified language and vision assets.	Supports repeatable local setup while remaining dependent on suitable host compute and the documented software environment.

The repository therefore demonstrates **software scalability through reuse and configuration**, not production-scale autonomous deployment. The next validation step is to measure how much time and customer-specific engineering are required to configure an additional site and robot.

1.7 Current Maturity and Evidence Boundary

The current result is an integrated simulation prototype. It demonstrates:

- natural-language mission decomposition;
- schema-constrained task planning;

- semantic-map resolution and deterministic route compilation;
- guarded navigation and independent human control;
- single-robot and four-robot simulated coordination;
- camera and LiDAR presentation;
- local visual perception and evidence qualification;
- findings-report generation and deployment history.

These are implemented software capabilities supported by the repository and the end-to-end demonstration. They must not be interpreted as evidence of certified functional safety, physical-robot reliability or customer return on investment.

A physical deployment would additionally require:

- a validated adapter for the physical robot and its safety interfaces;
- localization, mapping and sensor calibration at the deployment site;
- hardware emergency-stop integration and a site-specific risk assessment;
- customer-site perception data and target-specific qualification;
- cybersecurity, authentication, authorization and operational monitoring;
- maintenance, support and incident-response procedures.

This boundary is deliberate: the initiative demonstrates that an operational objective can be transformed into a constrained, executable and traceable ROS 2 mission in simulation. It does not claim that the current prototype is ready for unsupervised physical operation.

1.8 Success Definition and Next Validation Step

For the challenge, technical success means demonstrating that:

1. an operator can express a supported inspection objective in natural language;
2. the LLM decomposes the objective into predefined semantic tasks;
3. invalid or unsupported output is rejected before physical execution;
4. deterministic components resolve the route and invoke ROS 2 functionalities;
5. the simulated robot executes the mission while its progress remains visible and interruptible;
6. accepted observations retain the correct mission and pose identity;
7. the system produces the requested evidence and report;
8. completion, intervention and perception errors are measured using a documented protocol.

The recommended next step is a bounded warehouse pilot focused on one recurring inspection workflow, such as receiving-dock inspection. Before execution, the pilot should define the mission suite, expected targets, ground truth, success thresholds, intervention policy and responsible human sign-off.

The initiative's present value is therefore twofold. First, it provides a working demonstration of controlled LLM-to-ROS 2 mission execution. Second, it provides a reusable technical foundation on which a supervised warehouse inspection product can be measured, validated and progressively integrated with physical robotic systems.

Solution Architecture

2.1 Architectural Principle

The architecture translates a natural-language objective into a bounded and observable ROS 2 mission. Its central design principle is the separation of **semantic interpretation** from **physical execution**. The language model may propose what the robot should do, but deterministic components decide whether the proposal is valid and how it can be executed.

This separation addresses four requirements of the challenge:

- the operator can describe inspection, patrolling and monitoring objectives in natural language;
- the language model decomposes the objective using only predefined robotic tasks;
- semantic tasks are validated before they reach ROS 2 execution;
- stopping, manual intervention and safety decisions remain independent of generative inference.

The resulting architecture is deliberately restrictive. The language model cannot publish actuator commands, generate executable code, select arbitrary filesystem paths or bypass the configured map and safety constraints.

2.2 System Overview

The application is organized into presentation, interpretation, mission management, safety, execution and evidence layers. Figure 2.1 shows their logical relationships.

Browser communication uses local HTTP and WebSocket interfaces. Communication between the mission, perception and simulation processes uses ROS 2 topics and services. The portal can start one selected simulation runtime on demand and reuse the verified local language-model service across location changes. The fleet scenario instead starts four namespaced robot controllers in one shared MuJoCo world.

2.3 Component Responsibilities and Traceability

Table 2.1 connects each architectural layer to its responsibility and principal implementation. The listed paths are relative to the two packages under `src/`.

2.4 Mission Processing and Authority

A natural-language request is processed as a sequence of constrained transformations:

1. The browser submits the objective together with the active runtime and mission context.
2. Qwen3-4B, or the deterministic fallback, proposes a mission using the supported semantic task vocabulary.
3. The schema rejects unknown fields, invalid values, unsupported tasks and unresolved semantic locations.

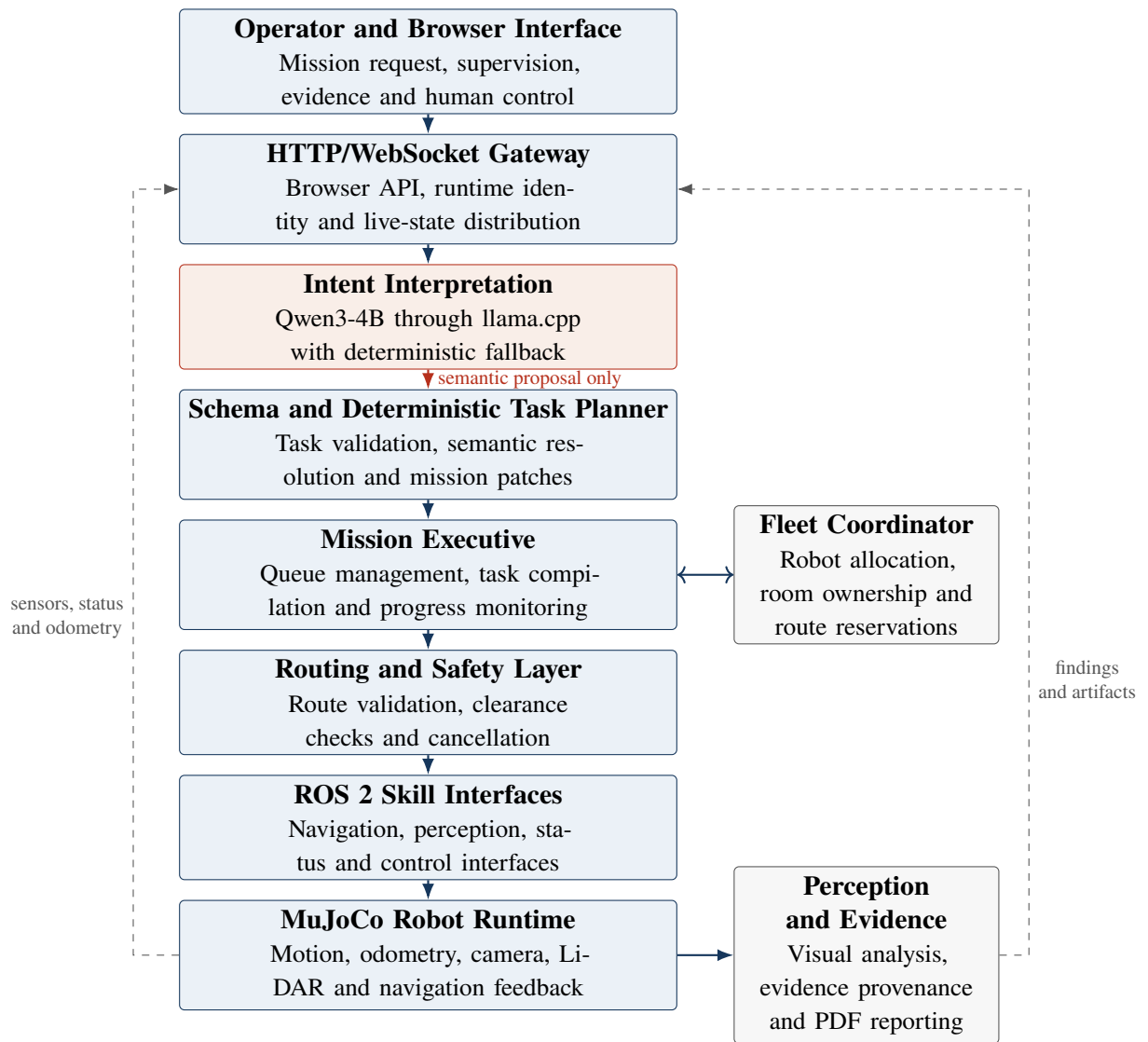


Figure 2.1: Logical architecture and authority boundary of the mission-control platform. The language model has no direct execution path to the robot.

Table 2.1: Responsibilities and implementation traceability of the principal architectural layers.

Layer	Responsibility	Principal implementation
Presentation and gateway	Accepts operator requests and presents mission state, cameras, LiDAR, evidence and independent human controls.	web/app.py, web_gateway_node.py
Intent interpretation	Converts natural language into a semantic mission proposal using the local model or bounded deterministic handling.	llm_planner.py, intent_parser.py
Schema and mission model	Restricts task types and fields, validates values and applies revision-checked mission modifications.	schema.py, mission.py
Executive and skill compiler	Converts accepted tasks into finite execution steps, manages their lifecycle and records outcomes.	task_execution.py, executive_node.py
Routing and safety	Resolves valid routes, checks clearance and prevents stale, blocked or unsupported goals from reaching execution.	planner.py, route_alternatives.py, navigation_safety.py
Fleet coordination	Allocates robots and coordinates rooms, routes, peers and charging resources without delegating arbitration to the language model.	fleet_coordinator_node.py, fleet_scheduler.py
Perception and evidence	Processes authorized visual requests and preserves evidence identity, qualification and report provenance.	perception_worker_node.py, perception_validation.py, reporting.py
Robot runtime	Simulates locomotion and publishes odometry, camera, LiDAR and navigation status.	simulator_node.py, fleet_simulator_node.py

4. A revision check prevents a delayed planning response from overwriting a newer operator instruction.
5. The mission executive compiles each accepted task into finite navigation, capture, inspection or reporting steps.
6. Deterministic routing and safety components validate physical goals before publishing them through ROS 2.
7. Runtime feedback updates task progress, while accepted evidence is associated with the active robot, mission, task, location, viewpoint, timestamp and pose.

The model may propose the following task categories:

- navigate_room and navigate_point;
- patrol_room;
- capture_room and inspect_room;
- return_station;
- report.

Named locations are converted into physical coordinates by deterministic map logic rather than by the language model. Likewise, route selection, fleet allocation, perception qualification and report authorization remain outside generative inference.

Stop, cancellation, manual takeover and upright recovery use separate control paths. They can therefore pre-empt an active mission without waiting for the language model or accepting a new plan.

2.5 Principal ROS 2 Interfaces

Table 2.2 lists the interfaces most relevant to the end-to-end mission path. JSON-structured application state is transported in bounded `std_msgs/msg/String` messages, while standard ROS 2 types are used for physical goals and sensor data.

Table 2.2: Principal ROS 2 interfaces in the mission execution path.

Interface	Type	Purpose
/go2/demo/command	<code>std_msgs/String</code>	Transfers the operator request from the gateway to the mission executive.
/go2/demo/state	<code>std_msgs/String</code>	Publishes the authoritative mission and task state.
/go2/nav/goal and /go2/nav/status	<code>PoseStamped</code> ; <code>String</code>	Transfer validated navigation goals and return navigation status.
/go2/demo/perception/request and /go2/demo/perception/result	<code>std_msgs/String</code>	Exchange bounded, identity-checked perception work and results.
/go2/odom, /go2/front_camera/image_raw and /go2/lidar/points	<code>Odometry</code> ; <code>Image</code> ; <code>PointCloud2</code>	Provide robot pose, visual observations and range measurements.
/go2/demo/stop, /go2/nav/cancel and /go2/recover_upright	Trigger services	Provide independent stopping, navigation cancellation and recovery.
/go2/demo/manual_override and /go2/manual/cmd_vel	<code>SetBool</code> ; <code>Twist</code>	Transfer and release control authority for bounded manual motion.

Fleet operation applies the same interface roles under robot-specific namespaces. This preserves per-robot state while allowing the fleet coordinator to maintain shared room and route constraints.

2.6 Safety and Failure Handling

The architecture fails closed when a mandatory precondition is not satisfied. Table 2.3 summarizes representative failure paths.

Table 2.3: Representative failure conditions and architectural responses.

Condition	Deterministic response	Result
Invalid model output	Reject unknown tasks, fields or out-of-range values.	No robot command is issued.
Ambiguous or unknown location	Require exact semantic-map resolution.	Clarification or bounded failure.
Stale plan or runtime identity	Compare mission revision and runtime generation.	Outdated work is rejected or quarantined.
Blocked or unsafe goal	Reject goals outside traversable space or without sufficient route clearance.	Mission remains stopped or attempts an eligible alternative.
Missing model or sensor	Apply readiness, freshness and timeout checks.	Dependent functionality remains unavailable.
Operator intervention	Invoke Stop, cancellation or manual ownership through the independent control path.	Active navigation is cancelled or held.
Fleet conflict	Enforce room ownership, route admission and peer-aware coordination.	Conflicting work waits or is reassigned.

Additional safeguards include bounded task and inference queues, finite retry budgets, evidence-identity validation and explicit authorization for persistent media or report generation. These controls constrain the demonstrated simulation behaviour; they do not constitute certified functional safety for a physical robot.

2.7 Design Rationale

The main architectural choices are justified as follows:

Constrained LLM authority

Natural language provides a flexible operator interface, while a strict schema and deterministic planner prevent unrestricted model output from controlling physical execution.

Local model execution

Repository-local language and vision models reduce dependence on external services and support deployments in which operational imagery should remain on-site.

ROS 2 modularity

Explicit interfaces separate interpretation, mission management, perception and robot execution, allowing components to be tested and replaced independently.

Simulation-first validation

MuJoCo provides a repeatable environment for testing mission logic, navigation and fleet coordination before introducing physical hardware and site-specific risk.

Independent human authority

Stop, cancellation, recovery and manual control remain available even when language inference is delayed, unavailable or incorrect.

2.8 Architectural Boundaries

The current architecture is a local simulation prototype operating within a trusted ROS 2 environment. It does not yet provide production authentication, remote multi-site fleet management, hardware emergency-stop integration or certified functional safety.

Physical deployment would additionally require a robot-specific control adapter, localization, calibrated sensors, communications monitoring, hardware safety interfaces and site-specific validation. These limitations are stated explicitly to distinguish the implemented software safeguards from claims about safe autonomous operation in a customer facility.

The contract of the principal inspection skill, including its inputs, preconditions, postconditions and safety constraints, is presented in [Chapter 3](#).

Documented and Working Robotic Skill

3.1 Skill Definition and Purpose

The principal skill documented for the challenge is `inspect_room`. It implements a bounded asset-inspection workflow that combines semantic mission interpretation, navigation, visual observation and evidence collection.

Its purpose is defined as follows:

Navigate through the configured inspection viewpoints of a semantic area, collect the requested visual observations and retain traceable evidence without allowing the language model to generate physical control commands.

The skill is composite rather than a monolithic ROS 2 node. Its behaviour is distributed across the task schema, semantic map, task compiler, mission executive, route planner, simulator and perception worker. This separation allows each stage to be validated independently.

The skill does not generate a findings report automatically. Reporting is a separate `report` task and requires an explicit operator request. This prevents an inspection request from implicitly authorizing a persistent filesystem export.

3.2 Skill Contract

Natural-language input is converted into a validated `TaskSpec`. The accepted fields are summarized in Table 3.1.

Table 3.1: Input contract of the configured-area inspection skill.

Field	Type	Constraint
<code>kind</code>	Constant	Must be <code>inspect_room</code> .
<code>room_id</code>	String	Must resolve to exactly one room in the active semantic map.
<code>queries</code>	List of strings	Optional visual targets such as <code>barrel</code> . At most 32 queries are accepted, with a maximum length of 160 characters per query.
<code>capture_interval_s</code>	Number	Accepted range of 0.2 s to 60 s; the default is 1 s.
<code>max_frames</code>	Integer	Accepted range of 1 to 256. This bound does not remove mandatory spatial coverage.
<code>save_images</code>	Boolean	Enables persistent image retention only when requested; the default is <code>false</code> .

The skill returns:

- a task status of `queued`, `running`, `cancelling`, `succeeded`, `failed` or `cancelled`;
- the compiled route and available planning diagnostics;
- structured observations for the authorized visual queries;
- evidence associated with the mission, robot, task, room, viewpoint, timestamp and measured pose;

- a completion summary or bounded failure reason.

Task success and perception success are intentionally different concepts. A task may complete every required navigation and capture step while producing an inconclusive visual observation. Conversely, an object is reported as a confirmed finding only when the active perception qualification permits that claim.

3.3 Preconditions

The skill can enter execution only when the following conditions hold:

1. an active runtime and validated semantic map exist for the selected location;
2. the requested room resolves unambiguously and contains at least one usable inspection pose or validated patrol seed;
3. current robot odometry is available and belongs to the active runtime generation;
4. a traversable route exists to every mandatory inspection viewpoint;
5. the front-camera and perception channels are ready when visual analysis is requested;
6. manual control or recovery does not own the robot and, in fleet mode, the coordinator has granted the required room and route admission.

If a mandatory precondition is not satisfied, physical execution is not dispatched. The task instead produces a clarification or bounded failure describing the rejected condition.

3.4 Execution Procedure

The accepted semantic task is compiled into a finite execution plan:

1. **Validate the task.** Reject unknown fields, unsupported values, excessive capture parameters and unresolved semantic references.
2. **Resolve inspection geometry.** Load the room and its configured inspection viewpoints from the semantic map. The language model cannot invent viewpoints or physical coordinates.
3. **Plan complete coverage.** Determine a collision-aware order through the required viewpoints and compile bounded navigation segments.
4. **Navigate to a viewpoint.** Publish the validated goal on `/go2/nav/goal` and monitor odometry, navigation status, obstruction, timeout and cancellation.
5. **Capture and analyse.** At the required pose and heading, publish an identity-bound perception request for the authorized queries.
6. **Validate the observation.** Accept the result only when its runtime, location, robot, task, room, camera, viewpoint and pose identities match the active request.
7. **Repeat and finalize.** Continue until every mandatory viewpoint has been processed, then clear active execution state and publish the terminal result.

This procedure ensures that an operator-specified image count cannot replace mandatory viewpoint coverage. It also prevents evidence from an earlier mission, robot or runtime from being attributed to the current inspection.

3.5 Skill State Model

Figure 3.1 presents the principal execution states. Stop or cancellation can pre-empt navigation and perception without waiting for language-model inference.

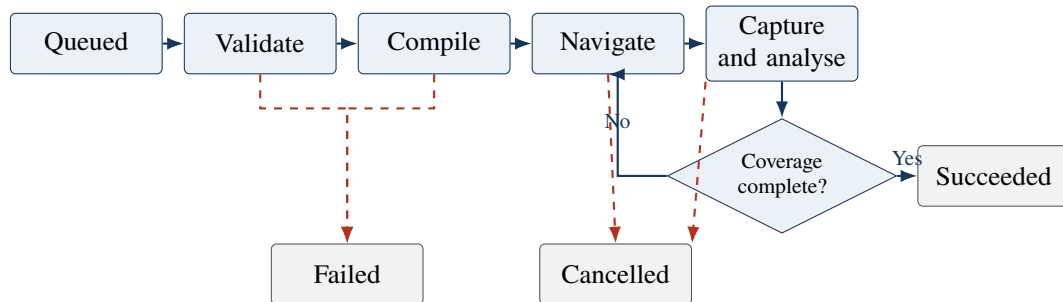


Figure 3.1: Simplified state model of the `inspect_room` skill. Dashed transitions represent rejection, failure or operator cancellation.

An unrecoverable failure during navigation, capture or perception also leads to `failed`. Temporary obstructions may instead invoke bounded replanning when an eligible alternative route exists.

3.6 Postconditions

Table 3.2 defines the terminal contract.

Table 3.2: Postconditions of the inspection skill.

Terminal state	Postconditions
succeeded	Every mandatory viewpoint has been processed; required headings and captures have completed; accepted evidence has the correct provenance; active goals and temporary execution state have been cleared.
failed	Execution has stopped; no additional views are requested; the failure reason is recorded; partial evidence remains explicitly incomplete and retains only its actual provenance.
cancelled	Active navigation has been cancelled or brought to a controlled hold; pending perception work is cancelled or quarantined; fleet resources are released when safe to do so.

No terminal state authorizes a findings PDF implicitly. A PDF is produced only if a separate, explicit `report` task satisfies its own authorization and evidence requirements.

3.7 Safety Constraints

The explicit safety constraints of the skill are listed in Table 3.3.

These mechanisms constrain behaviour within the simulation. They do not replace a hardware emergency stop, certified functional-safety system or site-specific risk assessment for a physical robot.

3.8 Implementation and Verification

The skill is traceable from its external contract to production code and automated verification, as shown in Table 3.4. Production paths are relative to `src/go2_ai_demo/go2_ai_demo/`; test paths

Table 3.3: Safety constraints of the `inspect_room` skill.

Constraint	Implemented behaviour
Constrained model authority	The language model selects semantic task fields only. It cannot generate actuator commands, executable code, inspection geometry or ROS 2 interfaces.
Semantic and route validity	Unknown rooms, goals outside traversable space and disconnected routes are rejected rather than approximated.
Complete spatial coverage	Capture preferences cannot remove mandatory inspection viewpoints.
Bounded execution	Query counts, frame counts, queues, retries and timeouts have explicit upper bounds.
Runtime and evidence identity	Stale commands or observations from another runtime, location, mission, robot or task are rejected or quarantined.
Human authority	Stop, cancellation, manual takeover and recovery can pre-empt the skill without consulting the language model.
Fleet admission	Room ownership and route coordination prevent conflicting fleet work from being dispatched simultaneously.
Evidence qualification	Unqualified, stale or spatially inconsistent observations cannot be presented as confirmed findings.
Explicit persistence	Permanent images and findings reports require explicit operator authorization.

are relative to `src/go2_ai_demo/test/`.

Table 3.4: Implementation and automated verification of the inspection skill.

Concern	Production implementation	Automated evidence
Contract and mission state	<code>schema.py</code> , <code>mission.py</code>	<code>test_schema_and_mission.py</code>
Viewpoint coverage and compilation	<code>task_execution.py</code> , <code>semantic_map.py</code>	<code>test_task_execution.py</code>
Execution and goal cleanup	<code>executive_node.py</code>	<code>test_navigation_stop_goal_cleanup.py</code> , <code>test_task_execution.py</code>
Perception qualification	<code>perception_worker_node.py</code> , <code>perception_validation.py</code>	<code>test_perception_validation.py</code> , <code>test_perception_verifier.py</code>
Evidence identity	<code>perception.py</code> , <code>web_gateway_node.py</code>	<code>test_evidence_identity_quarantine.py</code>
Cancellation and human control	<code>executive_node.py</code> , <code>web_gateway_node.py</code>	<code>test_runtime_stop.py</code> , <code>test_manual_control_gateway.py</code>
Evidence-based reporting	<code>reporting.py</code>	<code>test_reporting.py</code> , <code>test_report_progress.py</code>

The automated tests provide repeatable verification of individual contracts and failure paths. The end-to-end demonstration in [Chapter 4](#) provides the complementary integration evidence, while aggregate mission and perception results are reported in [Chapter 5](#).

3.9 Representative Execution

A representative operator instruction is:

“Inspect the receiving dock for barrels and save the inspection images.”

After semantic resolution, the validated task is equivalent to:

```
{
  "kind": "inspect_room",
  "room_id": "receiving_dock",
  "queries": ["barrel"],
  "capture_interval_s": 1.0,
  "max_frames": 12,
```

```
"save_images": true  
}
```

The resulting execution trace is:

```
request -> schema validation -> room resolution  
        -> viewpoint and route compilation  
        -> navigate -> capture -> validate evidence  
        -> repeat until coverage is complete  
        -> publish terminal task status
```

A successful execution confirms that the required movement and observation procedure completed. The presence of a barrel is reported separately as confirmed, absent, candidate or inconclusive according to the active perception qualification. This distinction prevents successful navigation from being misrepresented as successful object recognition.

End-to-End Demonstration

4.1 Choose an Operation

This chapter is a self-contained visual walkthrough of the running platform. The figures show what an operator sees and how a natural-language request moves from mission creation to robot execution, evidence review and reporting. No external media is required to understand the workflow.

After entering mission control, the operator first chooses the operational family. The default mission-library view presents two explicit choices: **Patrolling** for monitoring and inspection, and **Rescuing** for search and response. Each card explains its purpose, reports the number of configured locations and opens the corresponding deployment map. This prevents the operator from mixing site presets and workflows from different operational contexts.

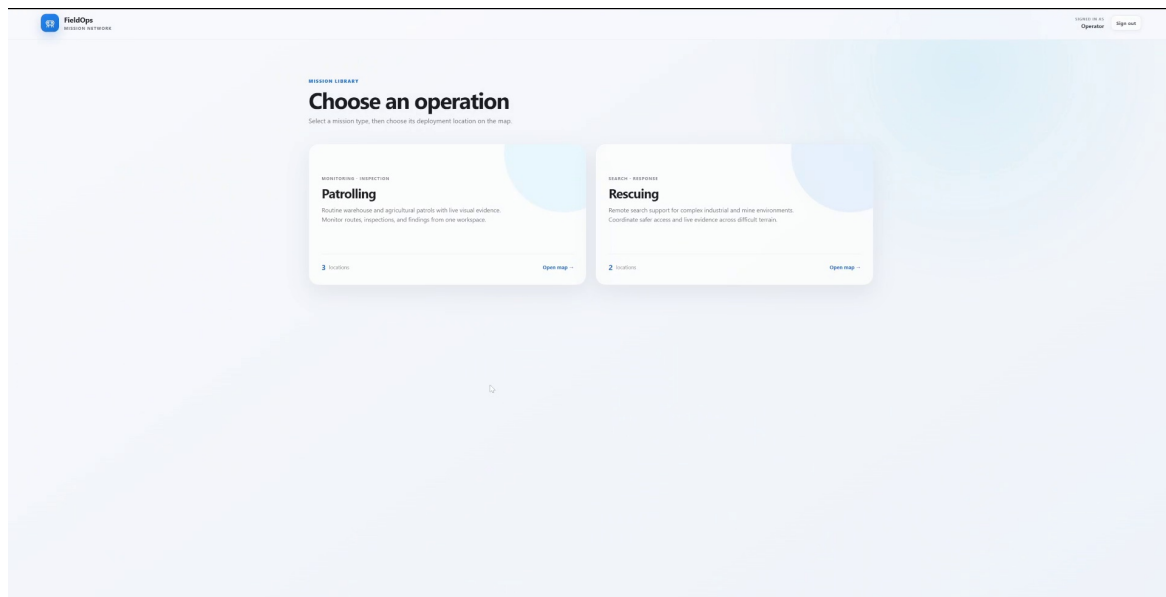


Figure 4.1: Default operation-selection screen. Patrolling and Rescuing are presented as separate mission families, with a short operational description, configured-location count and map entry point.

Moving over Patrolling gives immediate visual confirmation of the selected monitoring and inspection context before the operator opens its location map. The card describes routine warehouse and agricultural patrols, route monitoring, inspections and findings management.

Moving over Rescuing previews the search-and-response context. Its description sets a different operational expectation: remote search support, safer access and live evidence collection in complex industrial and mine environments.

4.2 Platform Workspace

The principal workspace is intentionally organized around one supervisory screen. The left column contains the mission conversation, live activity cards, the task board, evidence folders and the instruction box. The right side contains the spatial view, live LiDAR and robot camera. Navigation,

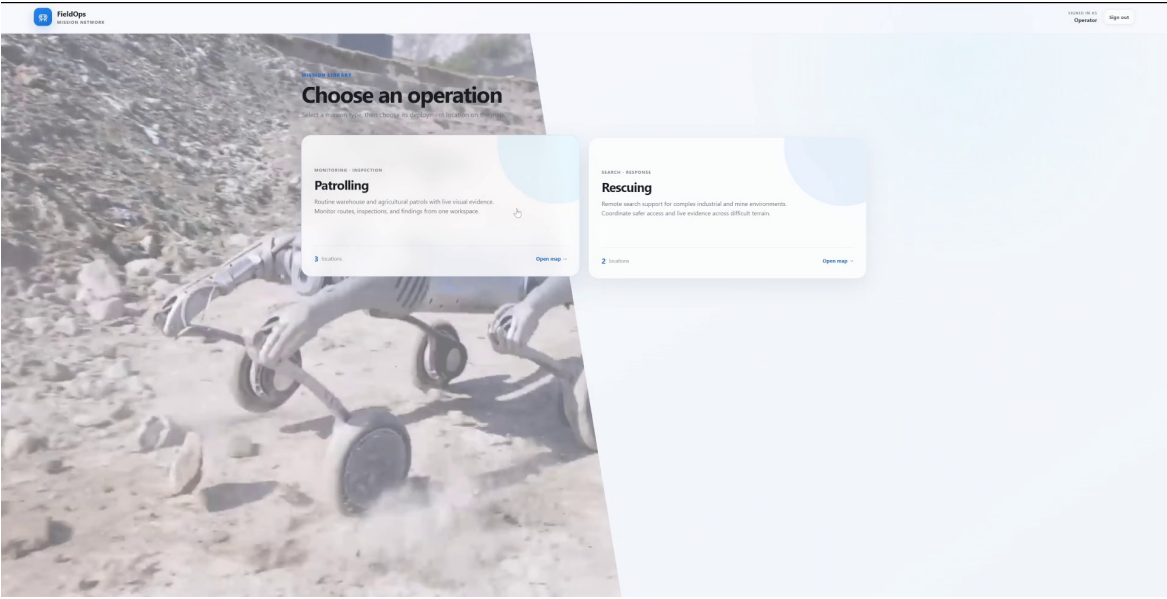


Figure 4.2: Patrolling operation preview. The background changes to the patrolling context while the card keeps its supported workflow and three configured locations visible.

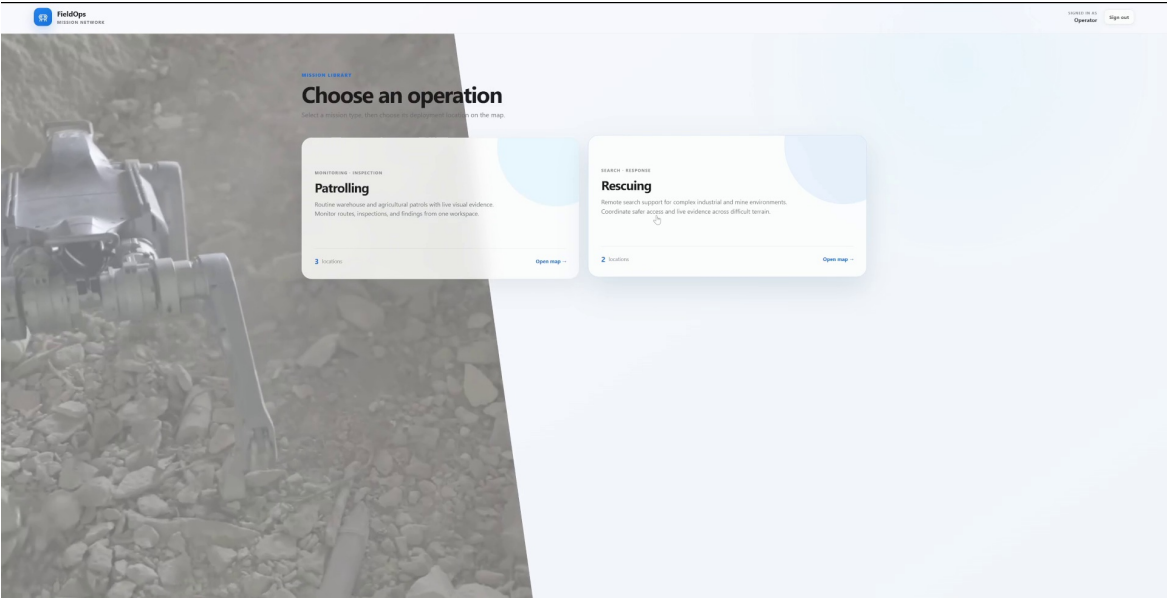


Figure 4.3: Rescuing operation preview. The search-and-response context is visually selected while the card identifies its two configured deployment locations and dedicated map entry point.

3D, plan and camera modes change the main view without hiding mission state, while the Stop control remains available during execution.

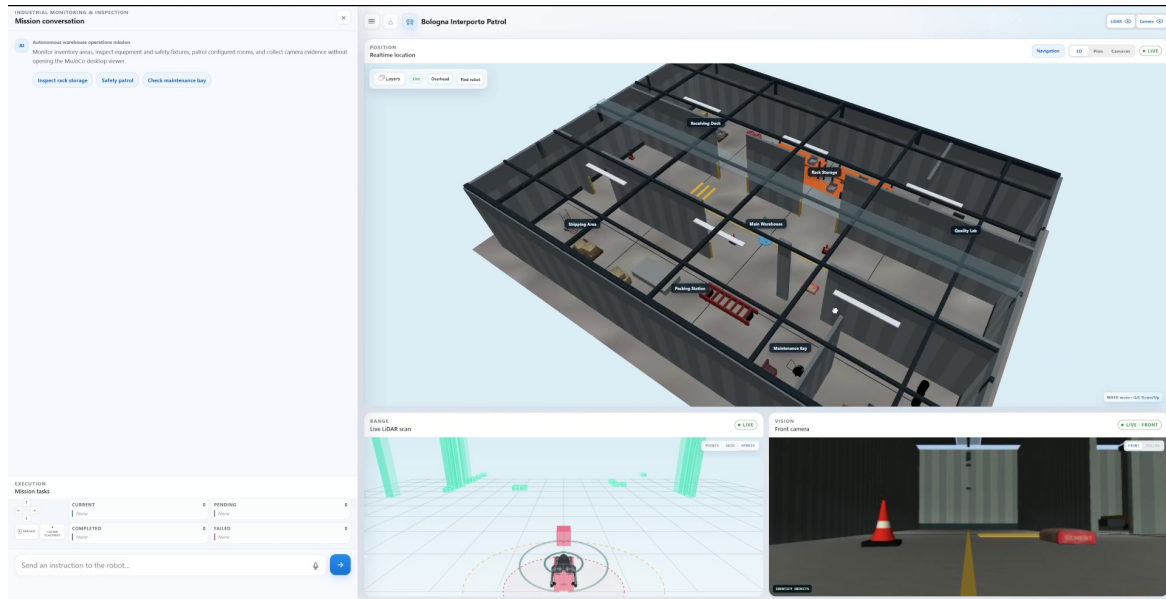


Figure 4.4: Single-robot operator workspace. Suggested missions and free-text input are on the left; the configured semantic environment, LiDAR and front camera are on the right; mission state and collected evidence remain visible below the conversation.

4.3 Create a Mission in Natural Language

The operator does not need to write ROS 2 commands or assemble a task graph. For the representative receiving-dock workflow, the operator enters:

“Go to the receiving dock, do an inspection and report it.”

The assistant resolves the configured location and expands the objective into three bounded tasks: navigate to the receiving dock, inspect the receiving dock and produce a findings report. The resulting queue is exposed to the operator; navigation is current, inspection and reporting are pending, and the planned route is drawn in the spatial view. The LLM therefore supplies semantic intent, while the task schema and mission executive determine what can execute.

4.4 Supervise Navigation and Inspection

Once accepted, the mission executive compiles the semantic tasks into guarded robot actions. The operator can follow the robot and route in 3D, switch to the topological plan, inspect the LiDAR safety envelope and observe the front camera. Progress cards distinguish navigation, inspection, analysis and report generation instead of presenting the mission as one opaque activity.

In Figure 4.6, navigation and inspection are complete, report generation is active, and the receiving-dock folder contains the captured observations. The workspace keeps the mission, task, room and evidence context on screen together, which helps the operator understand exactly what is running and where its data came from.

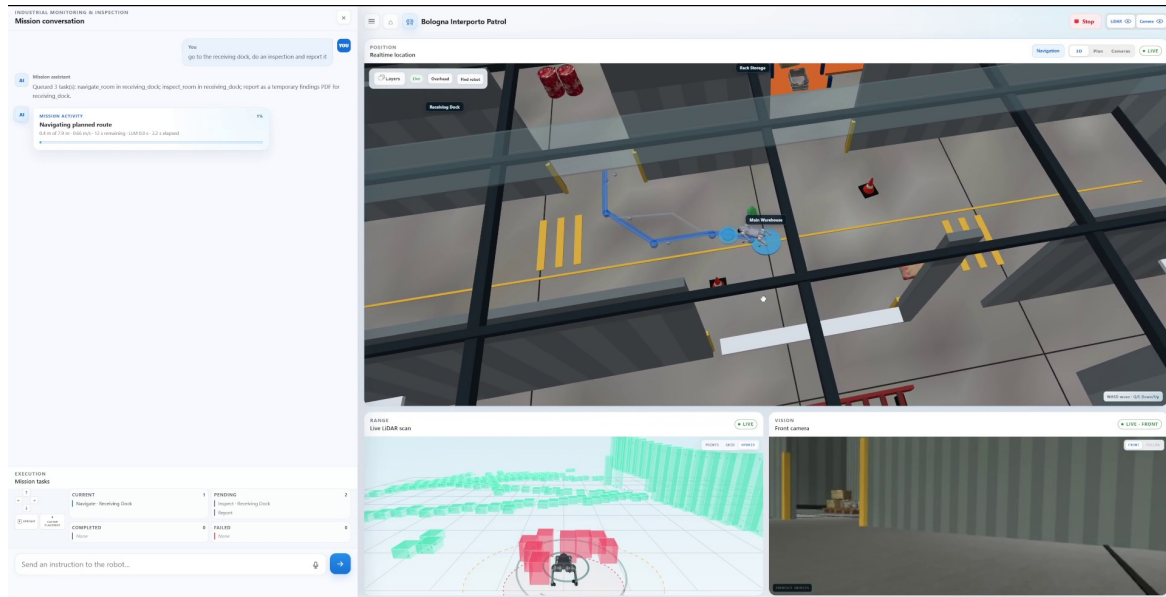


Figure 4.5: Natural-language mission creation. The original sentence, resolved task sequence, active navigation route, task-board state, live LiDAR, front camera and Stop control are visible together.

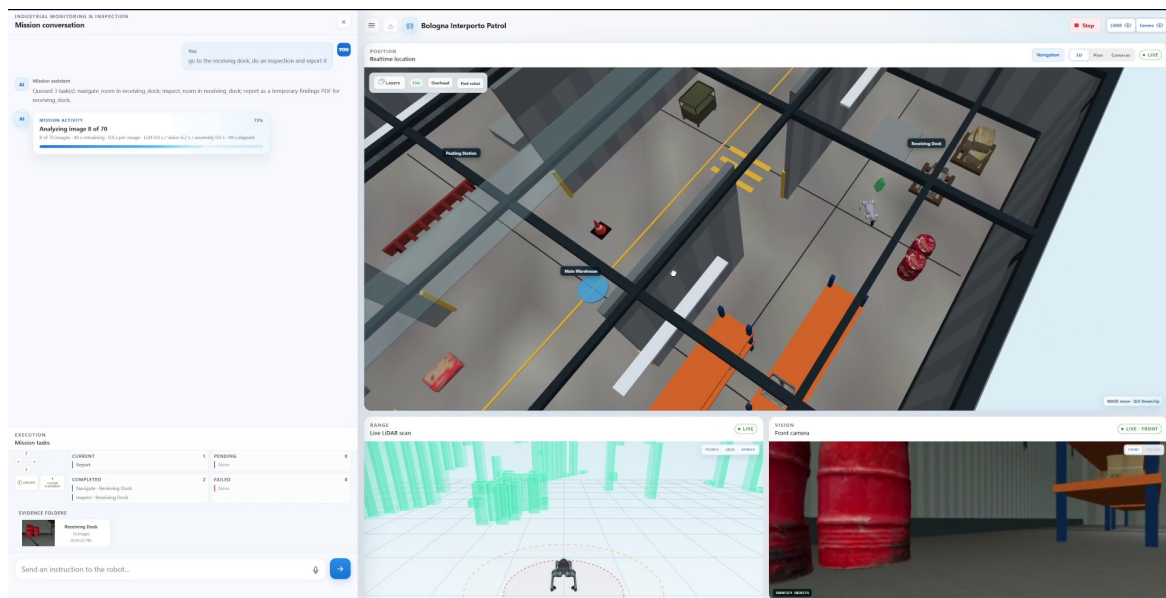


Figure 4.6: Live inspection and evidence processing. Completed and current tasks are explicit; the evidence folder is bound to the receiving dock; and the spatial, LiDAR and camera panes provide simultaneous execution context.

The Stop control is persistent during autonomous execution. It follows a dedicated cancellation path rather than being interpreted by the LLM, so the operator can pre-empt navigation or inspection even if mission reasoning or a perception request is still active.

4.5 Review Evidence, Not Only a Chat Answer

Selecting an evidence folder opens the evidence browser. Frames are grouped by task and room, retain capture times, and mark replayed keyframes. Unusable or empty frames are retained as part of the audit trail rather than silently removed; the operator can therefore distinguish what the robot captured from what the perception stack was able to use.

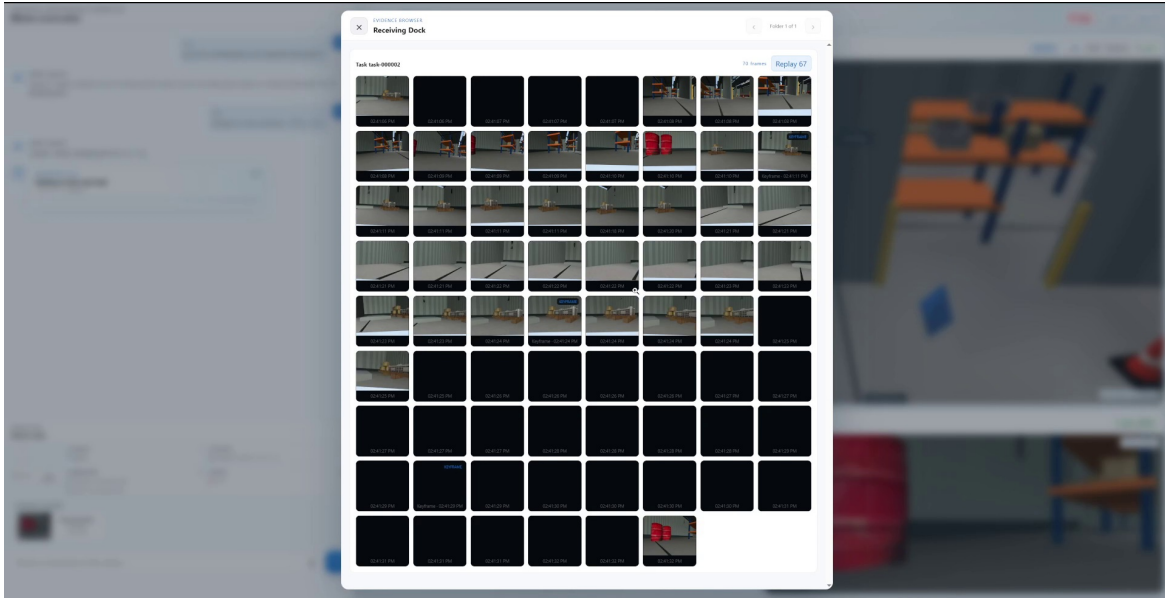


Figure 4.7: Receiving-dock evidence browser. Captured images are presented as one task-scoped collection with frame count, timestamps and keyframe/replay markers, preserving both useful and unusable observations.

Mission history remains visible when the operator submits another independent instruction. In Figure 4.8, the completed Navigate, Inspect and Report tasks, evidence folder, bounded perception conclusion and report-opening control remain available while a later navigation request is shown separately as current. This prevents results from being overwritten by the next command.

The generated findings PDF turns the captured evidence into an operator-ready artifact. Object sections show observation count, matching-frame count, maximum and mean confidence, representative images and source identity. Candidate-only outputs are labelled explicitly and are not presented as qualified presence claims.

4.6 Operate a Four-Robot Fleet

The same interface can supervise a fleet without creating four disconnected operator sessions. A single fleet objective is divided into non-duplicated room assignments. The fleet tab shows the shared state, while robot tabs expose each individual conversation and queue. Colour-coded routes, goals and robot markers make ownership visible in both the 2D plan and 3D views.

Fleet safety is also visible to the operator. When routes interact, the coordinator can move one robot to a configured clearance position and hold it before resuming the admitted route. The activity

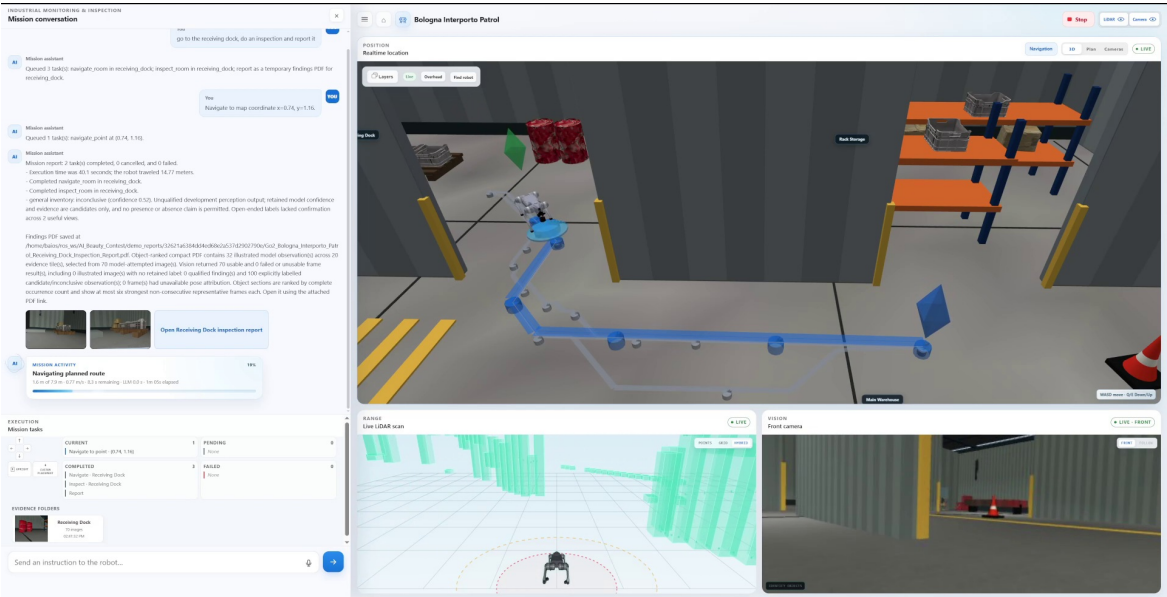


Figure 4.8: Persistent mission result and report hand-off. Completed tasks and zero failures remain in the task board; the conversation exposes execution details, the evidence qualification outcome, saved report path and a direct report-opening control.

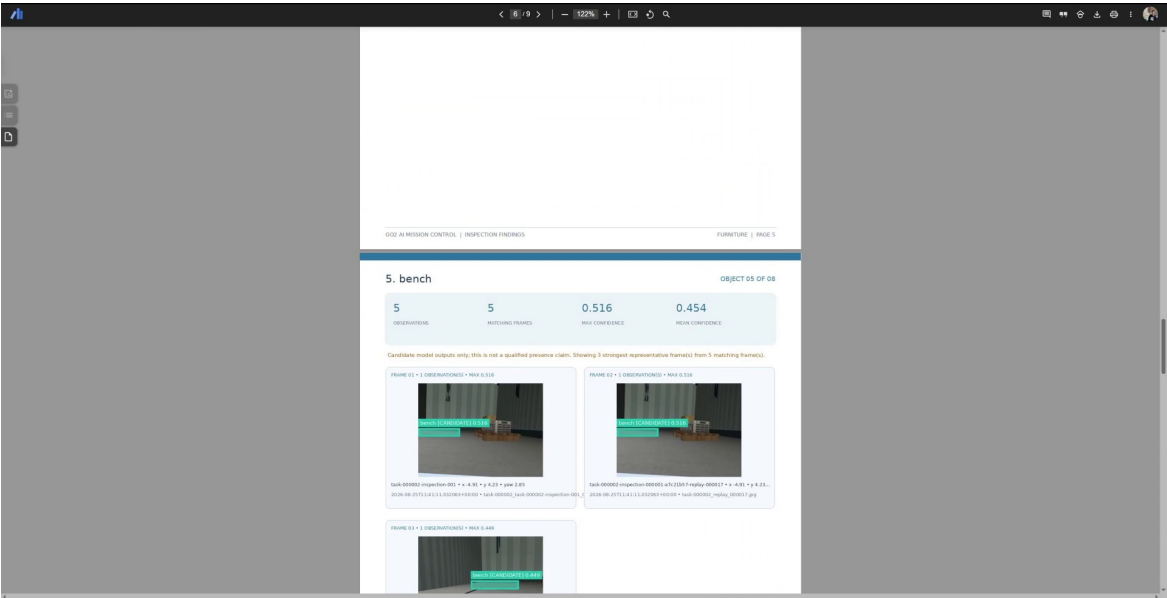


Figure 4.9: Evidence-based findings report. Each object section summarizes the supporting observations and confidence values and states when the result is only a candidate rather than a qualified presence claim.

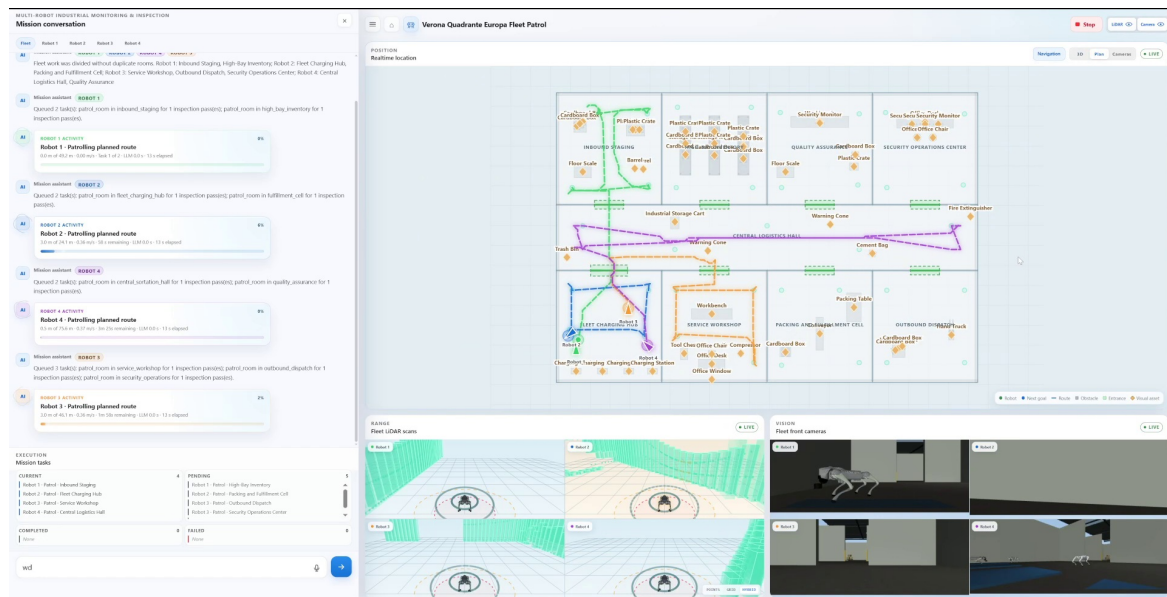


Figure 4.10: Fleet mission plan. Four task queues, colour-coded routes, assigned rooms, LiDAR panes and robot cameras are visible in one supervisory workspace.

stream reports this automatic route-clearing action, while the spatial view shows all robot positions and route ownership.

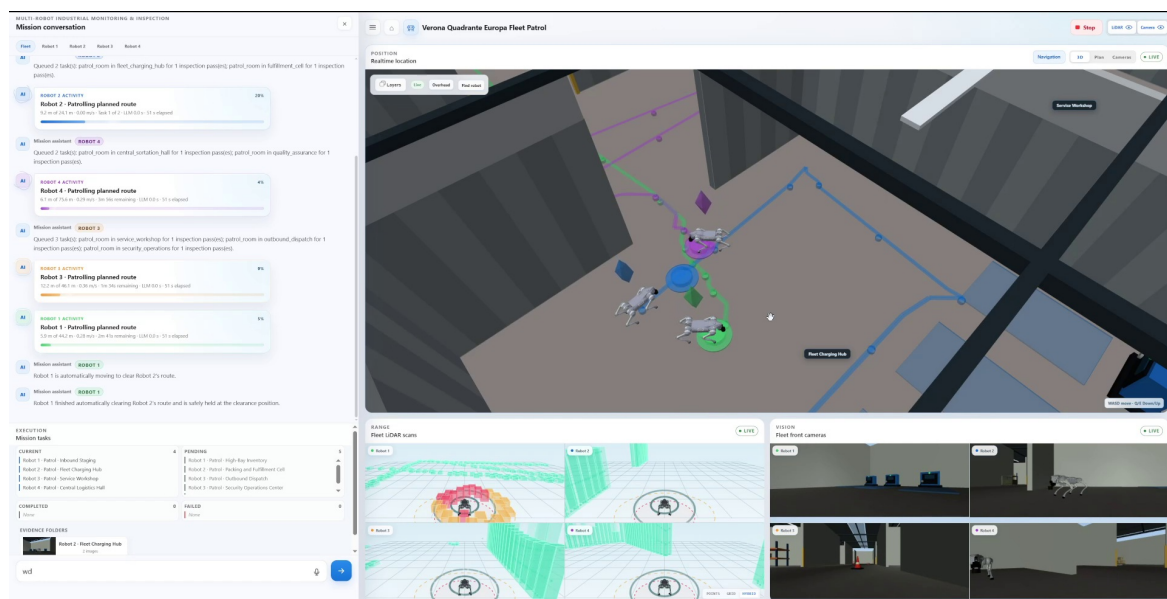


Figure 4.11: Automatic fleet coordination. The activity stream reports that Robot 1 clears Robot 2's route and reaches a safe hold position; the 3D view and per-robot sensor panes show the resulting spatial relationship.

The camera mode combines fixed environmental cameras with the four robot camera streams. Evidence folders stay accessible below the task board, allowing the operator to move directly from area-level awareness to the observations captured by a particular robot and room. Offline feeds are labelled rather than being replaced with stale imagery.

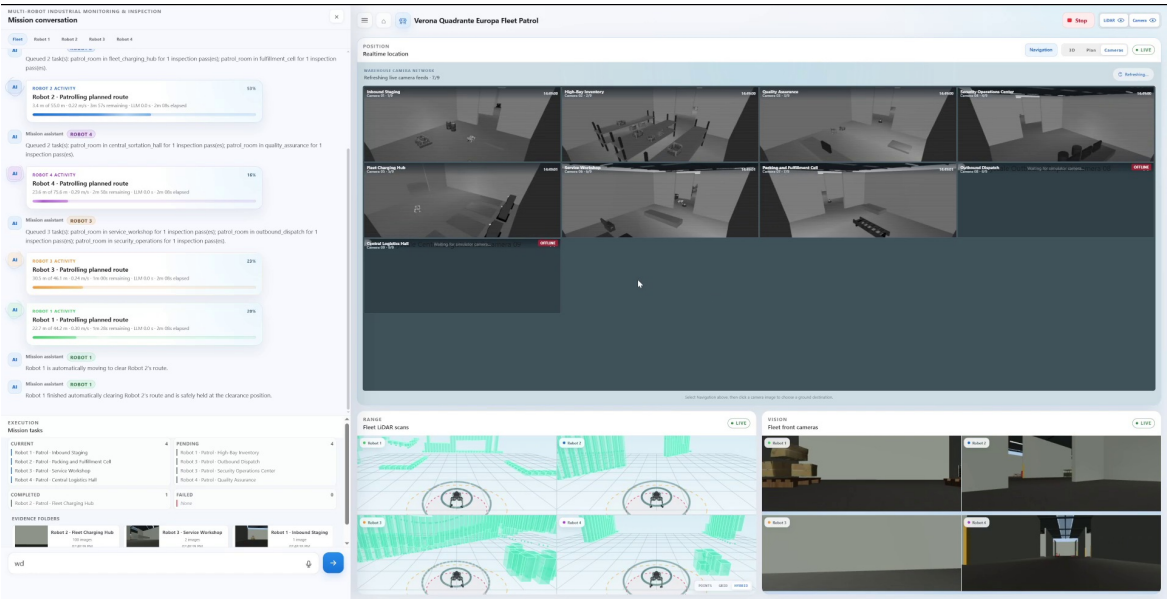


Figure 4.12: Fleet camera network and evidence overview. Fixed cameras, four robot cameras, task progress and room-scoped evidence folders are presented together; unavailable feeds are explicitly marked offline.

4.7 Select and Reuse Configured Deployments

The mission network separates reusable mission-control software from site-specific configuration. An operator first selects an available deployment from a map. The location card describes the mission, opens its workspace and shows previous deployment outcomes, including cancellations, so operational history is not hidden.

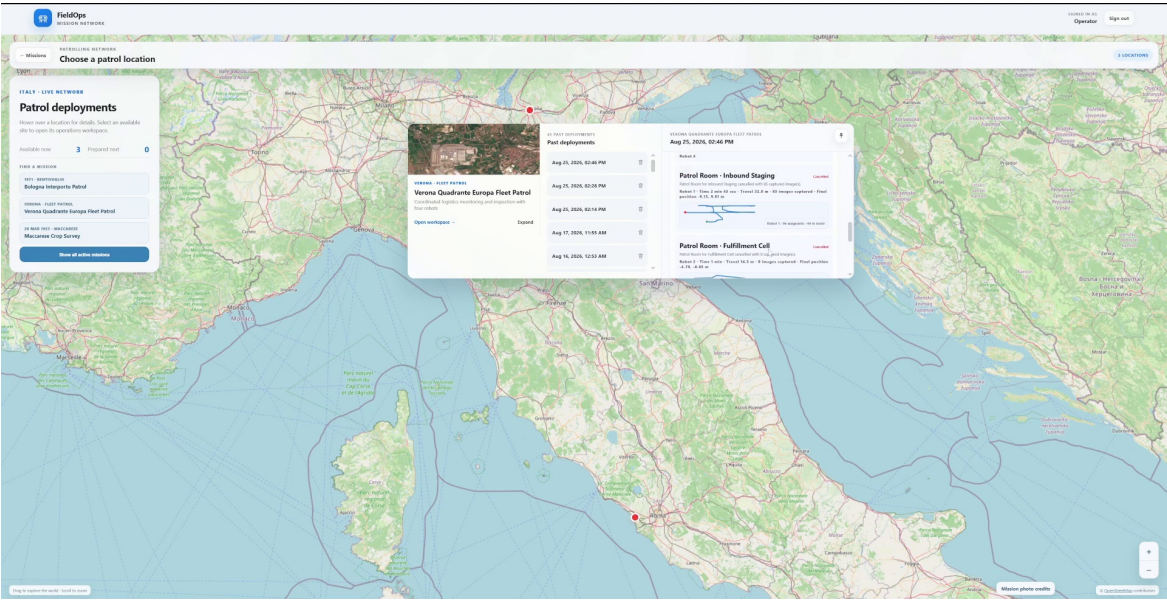


Figure 4.13: Deployment selection and history. The map presents available patrol locations, the selected site’s purpose, a workspace entry point and prior mission outcomes with route summaries.

After selection, the same operator model is applied to the site’s semantic map. The crop-survey configuration in Figure 4.14 includes a photogrammetric environment, named grid cells, automatically generated unsafe regions and a robot spawn location. These elements constrain later language requests without changing the core application.

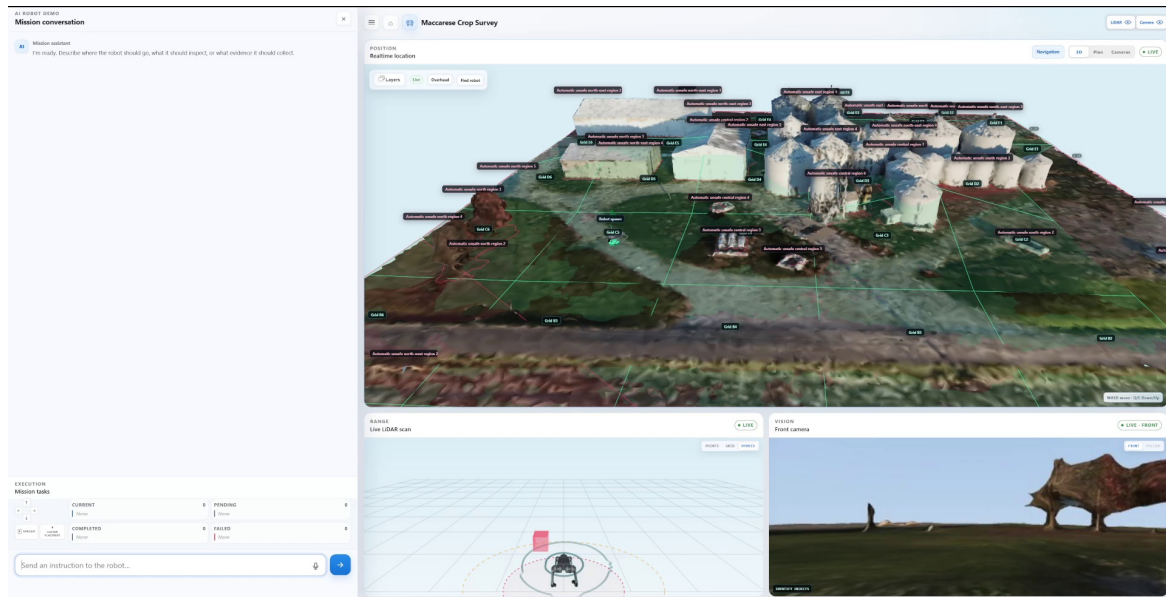


Figure 4.14: Configured crop-survey deployment. Site geometry, semantic grid, automatically derived unsafe regions, robot position, LiDAR and camera are loaded into the same mission workspace.

The real-site view in Figure 4.15 demonstrates the same navigation workflow on scanned terrain. A map-coordinate instruction is represented as a semantic task, the planned route is drawn on the reconstructed site, and the operator retains the same activity, task, LiDAR, camera and Stop controls used in the warehouse.

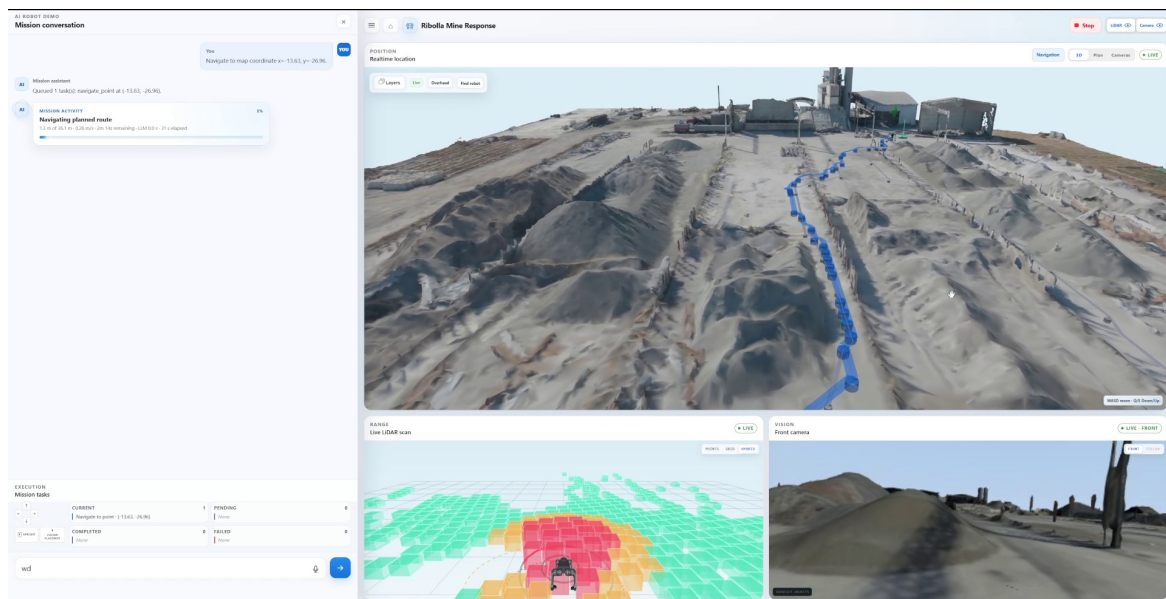


Figure 4.15: Navigation on a reconstructed real site. The route is displayed on the photogrammetric model while mission activity, task state, LiDAR, camera and operator authority remain unchanged.

4.8 End-to-End User Flow

The screenshots therefore document the complete usable loop:

operation selection → deployment selection → natural-language objective → validated

Table 4.1: How an operator uses the platform from operation selection to deliverable.

Step	Operator action	Platform response
1	Choose Patrolling or Rescuing.	Load the appropriate operational context and list of configured locations.
2	Select a configured deployment.	Load the site's semantic map, allowed areas, assets, sensors and mission history.
3	State the objective in natural language.	Resolve configured locations and assets, validate the proposal and expose the resulting semantic task queue.
4	Supervise execution.	Execute guarded ROS 2 actions while showing route, progress, LiDAR, camera and persistent Stop authority.
5	Review captured evidence.	Group observations by mission, robot, task and room while retaining keyframes and unusable-frame history.
6	Consume the result.	Publish terminal task state, bounded perception conclusions and a traceable findings PDF.
7	Scale supervision when required.	Allocate non-duplicated fleet work, coordinate route admission and combine robot and site-level sensing in one workspace.

semantic tasks → guarded ROS 2 execution → evidence review → bounded findings report.

This chapter demonstrates the integrated platform workflow. Aggregate mission completion, human-intervention and perception rates remain separate measurements and are reported only in [Chapter 5](#).

KPIs and Results

5.1 Evaluation Scope and Evidence Source

The results in this chapter are derived from the automatically generated evidence packages checked into `kpi_results/`. The recorder observes ROS 2 mission, navigation, control, perception and artifact events without changing the application under evaluation. Each package retains its frozen environment, raw event log, mission rows, intervention records, perception cases, copied artifacts and a hash manifest. Failed, cancelled, timed-out and incomplete runs remain in the evidence and are not removed from a denominator.

At the time of this report, the repository contains 11 timestamped packages created on 25 August 2026. They contain six recorded operator instructions: two formally classified nominal trials and four successful but unclassified navigation instructions. Only the two nominal trials are eligible for the mission-completion denominator. Empty packages and unclassified instructions are retained as operational evidence but are not counted as nominal successes or failures.

Both nominal packages were produced at Git commit `9f065b24f8082dedc1baf0bafb7968305b76584a` on Ubuntu 22.04.5 under WSL 2, ROS 2 Humble, Python 3.10.12 and MuJoCo 3.10.0. The host used an AMD Ryzen 7 3700X processor, 16.7 GB of RAM and an NVIDIA GeForce RTX 3070 with 8 GB of memory. The repository working tree was not clean, so the recorded commit alone cannot reconstruct the exact evaluated source. The single-robot and fleet trials also have distinct configuration group identifiers and are therefore reported separately rather than merged into one frozen-configuration result.

The single-robot package `kpi_results/2026-08-25_1440/` used the `warehouse_ai_demo_v0` map, the PGTT locomotion profile, guarded navigation, automatic instruction planning and local candidate-only perception. The fleet package `kpi_results/2026-08-25_1446/` used the `verona_fleet_warehouse_v0` map with four simulated Go2 robots. Neither package had a passing perception-qualification artifact.

5.2 KPI Definitions and Denominator Rules

A nominal mission is successful only if every requested semantic task reaches succeeded, every mandatory inspection viewpoint is processed, every explicitly requested permanent image and report is produced, and no requested task remains unresolved. A ROS 2 mission state of `succeeded` is therefore necessary but not sufficient for KPI success: the recorded task multiset and requested outputs must also match the declared scenario.

The principal challenge KPIs are calculated as follows:

$$\text{Mission completion rate} = \frac{\text{successful nominal missions}}{\text{attempted nominal missions}} \times 100, \quad (5.1)$$

$$\text{Human intervention rate} = \frac{\text{nominal missions requiring an unplanned intervention}}{\text{attempted nominal missions}} \times 100, \quad (5.2)$$

$$\text{False-positive rate} = \frac{FP}{FP + TN} \times 100, \quad (5.3)$$

$$\text{False-negative rate} = \frac{FN}{FN + TP} \times 100. \quad (5.4)$$

Planned Stop, manual-control, recovery and rejection demonstrations are safety tests and are excluded from the nominal mission and human-intervention denominators. Unclassified instructions are also excluded. Perception values may enter the confusion matrix only when they come from a separately labelled, held-out final-evaluation set with a matching passing qualification artifact. Candidate detections and semantic-map inventory are not ground truth.

Secondary measures include evidence coverage, report-generation success, mission duration, report-availability latency, navigation replans and recoveries, Stop acknowledgement latency and fleet conflict-free completion. Missing or causally ambiguous observations remain incomplete; they are never converted into zero.

5.3 Planned Nominal Mission Matrix

The documented evaluation protocol requires at least five executions of each of five scenarios under a frozen configuration, for a total of 25 nominal trials. Table 5.1 compares that target with the eligible records currently available.

Table 5.1: Recorded nominal trials compared with the evaluation protocol.

Scenario	Recorded	Required	Shortfall
Receiving Dock barrels with saved images	0	5	5
Maintenance Bay equipment inspection	0	5	5
Complete warehouse patrol	0	5	5
Receiving Dock inspection with findings PDF	1	5	4
Four-robot fleet inspection	1	5	4
Total	2	25	23

The nominal suite is consequently incomplete. The values below describe the two retained trials and must not be interpreted as statistically representative mission reliability.

5.4 Recorded Mission Results

Table 5.2 reports each nominal trial exactly as classified by its generated package. The status column is the application terminal state; the KPI column applies the stricter scenario-completion contract.

For configuration group `config-9b673d9835ab`, the observed mission completion rate is (0/1=0%). The application completed the receiving-dock navigation, all three inspection viewpoints and the requested report, but the strict evaluator rejected the trial at semantic-task validation because the executed plan contained an additional `navigate_room` task beyond the declared scenario task set. The run is therefore retained as a failed nominal trial even though its application terminal state was

Table 5.2: Recorded nominal mission results.

Package	Scenario	Terminal state	KPI success	Duration	Coverage and output
1440	Receiving Dock inspection with PDF	Succeeded	No	117.645 s	3/3 viewpoints, three evidence records and one requested PDF.
1446	Four-robot fleet inspection	Cancelled	No	179.066 s	12/20 viewpoints and 12 evidence records before cancel-and-hold.

succeeded.

For configuration group `config-0ec20671d5c9`, the observed mission completion rate is also (0/1=0%. The four-robot mission was cancelled after 12 of 20 required viewpoints had been completed. The simulator acknowledged a cancel-and-hold, and the incomplete work remains in the denominator. Combining these two percentages into a single formal completion rate would violate the frozen-configuration rule; descriptively, neither of the two recorded nominal trials met the complete scenario contract.

Four additional navigation instructions reached succeeded, including three coordinate goals and one named real-location grid goal. Because they were not declared as one of the five nominal scenarios, they provide supplementary end-to-end navigation evidence only and do not improve the official completion rate.

5.5 Human Intervention

The numeric human-intervention KPI is **incomplete**. The fleet trial contains one observed control episode associated with its cancellation, while the single-robot nominal row records no intervention. However, the current dashboard does not authoritatively distinguish a new instruction from a human correction or resubmission, and the retained intervention records include unresolved classification or outcome fields. The package-wide summary therefore correctly emits a null intervention rate rather than reporting either (0%, (50% or (100%.

The fleet package measured a 22.510 ms Stop service acknowledgement from gateway handler entry to acknowledgement by the targeted executive services. This excludes browser and network dispatch as well as physical settling, and the action was not registered as a planned safety trial. It is consequently a diagnostic latency observation, not a completed safety-test KPI.

5.6 Perception False Positives and False Negatives

False-positive and false-negative rates are **not yet measurable** from the checked-in packages. Every `perception_cases.csv` contains only its schema header: no independently labelled held-out case was imported. The runtime used real local inference with unqualified results allowed for development display, but no class had a matching passing qualification artifact. Visible boxes, masks, confidences and candidate observations are therefore evidence of model execution, not confirmed presence or absence.

Reporting these unavailable values as (0% would incorrectly imply that no classification errors occurred. A final measurement requires a sealed held-out dataset, a proven calibration/final split and target-specific ground truth for both positive and negative cases.

Table 5.3: Current status of the required perception KPIs.

Measure	Result	Reason
True/false positive counts	Unavailable	No sealed, independently labelled final-evaluation cases.
True/false negative counts	Unavailable	No qualified presence or absence decisions may be scored.
False-positive rate	Unavailable	The denominator (FP+TN) is not established.
False-negative rate	Unavailable	The denominator (FN+TP) is not established.
Classification coverage	Unavailable	Candidate-only outputs are not decisive qualified classifications.

5.7 Secondary Observations

The single-robot run achieved (3/3=100% viewpoint coverage and generated the one requested findings report. The recorder observed the report as already durable when the terminal mission state arrived, yielding a recorded terminal-to-durable latency of 0 s. This boundary does not measure the full report-generation time from the beginning of the report task.

The cancelled fleet run achieved (12/20=60% viewpoint coverage. Its fleet observer retained 12,073 fresh samples and observed no simulator peer-contact-emergency episode, but it also recorded two coverage gaps and a maximum pose skew of approximately 0.237 s. Fleet conflict-free completion therefore remains incomplete. Both nominal summaries report zero counted navigation recoveries. The fleet log contains one observed replan transition even though the finalized aggregate replan counter is zero; both values are retained rather than silently reconciled.

No intentional safety or rejection tests, instruction-regression import, or the six required browser screenshot categories are present in the packages. The generated reproduction scripts recreate the application launch, but not the sequence of 25 nominal trials, induced safety cases, labelled perception imports or screenshot timing.

5.8 Result Status and Next Measurement Gate

Table 5.4 summarizes what can and cannot currently be claimed.

Table 5.4: KPI measurement status at the report date.

KPI	Status	Current evidence
Mission completion rate	Preliminary	Two nominal trials in separate configuration groups; both failed the complete scenario contract.
Human intervention rate	Incomplete	Intervention classification and correction/resubmission tracking are not complete.
False-positive rate	Incomplete	No independently labelled held-out negative cases.
False-negative rate	Incomplete	No independently labelled held-out positive cases.
Evidence coverage	Preliminary	100% in one single-robot trial and 60% in one cancelled fleet trial.
Report-generation success	Preliminary	One requested report was generated; sample size one.

The next acceptance gate is to execute all five nominal scenarios at least five times under documented frozen configurations, complete the separate intentional safety suite, import a sealed held-out perception evaluation and capture the required browser evidence. Human corrections and resubmissions must

also receive explicit evaluator identities so that the intervention rate has an auditable numerator.

The present results prove that the KPI recorder can retain mission outcomes, coverage, artifacts and failures without discarding adverse runs. They do not yet establish a reliable mission-completion rate, human-intervention rate or perception error rate. All observations apply only to the documented MuJoCo simulation configurations and do not establish physical-robot reliability, certified functional safety or customer-site performance.

Business Case

6.1 Scope and Commercial Recommendation

The current application is an integrated simulation prototype rather than a production-ready physical inspection system. The initial commercial offer should therefore remain close to the demonstrated level of maturity.

The proposed first offer is a **paid simulation-first warehouse inspection pilot**. During the pilot, the customer's inspection workflow is represented in a configured simulation, translated into bounded robotic missions and evaluated using agreed technical and operational KPIs. The pilot produces a validated mission workflow, an operator demonstration, inspection evidence and a plan for subsequent physical integration.

Commercial recommendation: offer a bounded warehouse-inspection pilot as an engineering and validation service, followed by site onboarding and an annual mission-control software licence when physical deployment becomes technically and commercially justified.

This approach generates initial service revenue while allowing the reusable software platform to develop into a more scalable licence-based product. It also avoids presenting simulation results as evidence of physical-robot readiness.

6.2 Initial Customer and Operational Problem

The initial target customer is a warehouse or logistics operator that performs frequent visual inspections and is already evaluating mobile robotics or working with a robotics systems integrator. A suitable first customer has:

- one warehouse with clearly defined inspection areas;
- recurring and documentable inspection workflows;
- an operations or safety manager responsible for the process;
- an automation team or integration partner;
- a willingness to compare the proposed workflow with the current manual procedure.

The first use case is a recurring inspection of areas such as receiving, storage and shipping. The robot is requested to visit configured viewpoints, collect visual evidence and produce a findings report when explicitly requested.

The expected value is not based on replacing the complete human inspection process. It is based on reducing routine operator effort, improving repeatable coverage and simplifying the association of evidence with a mission, location and time.

6.3 Proposed Commercial Offer

The pilot is intentionally limited to one site and a small number of inspection workflows. This limits delivery risk and makes the result measurable.

Table 6.1: Proposed scope of the initial commercial pilot.

Element	Proposed scope
Duration	Eight to twelve weeks.
Site	One warehouse or representative warehouse model.
Workflows	Up to three inspection, patrolling or monitoring missions.
Configuration	Semantic areas, inspection viewpoints, authorized visual targets and reporting rules.
Demonstration	Natural-language mission creation, supervised execution, evidence review and findings-report generation.
Validation	Agreed mission-completion, intervention and perception KPIs.
Training	Two operator training and review sessions.
Exclusions	Physical-robot certification, production safety approval, enterprise integration and hardware procurement.

The pilot should conclude with a decision on whether to stop, repeat the validation, proceed to physical integration or prepare a limited production deployment.

6.4 Cost and Pricing Assumptions

No customer quotations or measured delivery costs are currently available. Consequently, the following values are **planning assumptions** used to construct an initial financial case. They are not presented as market-validated prices.

The model assumes a fully loaded project labour rate of **EUR 50 per hour**. Hardware purchase is excluded because the first pilot is simulation-first. A physical robot, additional sensors and on-site compute would require a separate quotation.

Table 6.2: Estimated direct cost of one simulation-first pilot.

Cost item	Estimated cost
Workflow discovery and requirements, 40 hours	EUR 2,000
Site and mission configuration, 100 hours	EUR 5,000
Testing and KPI measurement, 80 hours	EUR 4,000
Documentation and operator training, 24 hours	EUR 1,200
Project management and pilot support, 36 hours	EUR 1,800
Compute, travel and pilot materials	EUR 2,000
Contingency, 15% of direct cost	EUR 2,400
Estimated pilot delivery cost	EUR 18,400

A proposed pilot price of **EUR 25,000** produces an estimated direct contribution of:

$$\text{EUR } 25,000 - \text{EUR } 18,400 = \text{EUR } 6,600. \quad (6.1)$$

The corresponding contribution margin is approximately 26%. This margin is intentionally moderate because early pilots contain significant customer-specific engineering.

If a pilot progresses to a limited production deployment, the proposed commercial structure is shown in Table 6.3.

Table 6.3: Initial pricing and direct-cost assumptions.

Commercial item	Revenue	Direct cost	Contribution
Simulation-first pilot	EUR 25,000	EUR 18,400	EUR 6,600
Production-site onboarding	EUR 8,000	EUR 4,000	EUR 4,000
Annual software licence	EUR 12,000	EUR 2,000	EUR 10,000
Annual support package	EUR 4,000	EUR 3,000	EUR 1,000

The annual recurring package therefore produces EUR 16,000 of revenue and an estimated EUR 11,000 contribution per site before fixed development costs, general administration and tax. Robot hardware, maintenance, insurance and physical integration are excluded and must be supplied or priced separately.

6.5 Expected Revenue Scenarios

Table 6.4 presents three illustrative first-year scenarios. A conversion represents a pilot customer that purchases site onboarding and one annual software and support package. The calculation assumes that converted customers are invoiced early enough for the first annual package to be recognized within the same year.

Table 6.4: Illustrative first-year revenue scenarios.

Measure	Conservative	Base	Upside
Paid pilots	1	3	6
Production conversions	0	1	3
Expected revenue	EUR 25,000	EUR 99,000	EUR 222,000
Estimated direct cost	EUR 18,400	EUR 64,200	EUR 137,400
Estimated contribution	EUR 6,600	EUR 34,800	EUR 84,600
Contribution margin	26%	35%	38%
Year-end recurring revenue	EUR 0	EUR 16,000	EUR 48,000

These scenarios are not sales forecasts. They show the financial consequences of a small number of transparent assumptions. Customer interviews and priced pilot discussions are required before the conversion rates and achievable prices can be considered reliable.

6.6 Illustrative Customer Value

The customer case should be evaluated against the existing inspection process. For illustration, consider a warehouse performing four inspection and documentation rounds per day over 250 working days. If each round requires 1.25 hours at a fully loaded labour cost of EUR 35 per hour, the annual baseline cost is:

$$4 \times 250 \times 1.25 \times \text{EUR } 35 = \text{EUR } 43,750. \quad (6.2)$$

If supervised robotic execution reduces active human time by 50%, the potential annual labour saving is EUR 21,875. After an annual software and support cost of EUR 16,000, the remaining measurable annual benefit is EUR 5,875.

Under these assumptions, the EUR 8,000 production-site onboarding fee has a simple payback period of approximately 16 months:

$$\frac{\text{EUR } 8,000}{\text{EUR } 5,875/12} \approx 16.3 \text{ months.} \quad (6.3)$$

This calculation excludes robot purchase, leasing and physical integration. It is therefore most relevant to a customer that already owns a compatible robot or obtains it through an integration partner. It also excludes possible benefits from increased coverage, improved evidence quality, reduced exposure to hazardous areas and avoided downtime because these benefits have not yet been measured.

6.7 Scalability

Early deployments will require substantial engineering effort. The commercial model becomes more scalable only if customer-specific work is progressively replaced by reusable software and standard procedures.

The main scaling mechanisms are:

- configuration-driven semantic maps and inspection viewpoints;
- reusable mission and skill contracts;
- standard warehouse mission templates;
- repository-local language and perception models;
- repeatable site-onboarding and acceptance procedures;
- annual site licences and standardized support packages;
- delivery through robotics and systems-integration partners.

The principal scaling risk is excessive customization. If every customer requires a different robot interface, map format, perception model and reporting workflow, the business remains an engineering service with limited margins. A pilot should therefore measure configuration and support effort as carefully as mission performance.

6.8 Risks and Decision Gates

6.9 Business-Case Conclusion

The recommended first commercial step is a EUR 25,000 simulation-first warehouse-inspection pilot. Under the stated assumptions, the pilot costs approximately EUR 18,400 to deliver and produces a direct contribution of EUR 6,600. A successful pilot can progress to EUR 8,000 of production-site onboarding and EUR 16,000 of annual recurring software and support revenue.

The base first-year scenario consists of three paid pilots and one production conversion, producing an estimated EUR 99,000 in revenue and EUR 34,800 in direct contribution. These figures are planning assumptions rather than validated forecasts.

The business case is attractive only if three conditions are demonstrated: customers are willing to pay for the pilot, customer-specific delivery effort decreases across sites, and physical operation can meet the agreed safety, mission and perception requirements. The next commercial action is therefore to present the bounded pilot to suitable warehouse operators or integration partners and replace the assumptions in this chapter with measured quotations, delivery hours and customer baseline data.

Table 6.5: Principal business risks and corresponding decision gates.

Risk	Evidence required	Decision gate
Insufficient customer need	Interviews and measurement of the current inspection workflow.	Do not start a pilot without a frequent and measurable problem.
Low willingness to pay	A priced proposal reviewed by a qualified customer.	Proceed only with a paid pilot or an explicitly funded validation.
Excessive delivery effort	Actual hours spent on mapping, configuration, testing and support.	Continue only if later sites can reuse a substantial part of the work.
Physical integration risk	Robot interface, localization, sensor and emergency-stop validation.	Do not claim production readiness before physical acceptance testing.
Unqualified perception	Customer-site ground truth and measured false-positive and false-negative results.	Keep findings advisory until the required class passes qualification.
Licensing and security gaps	Review of software, models, assets, data handling and deployment security.	Resolve material issues before external commercial deployment.